Fiscal Year:	FY 2012	**Task Last Updated:**	FY 08/02/2011
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Project Title:	Assessing and Mitigating the Impact of Transmission Delays on Teleoperations		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Space Human Factors Engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SHFH**:Space Human Factors & Habitability (archival in 2017)		
Human Research Program Risks:	(1) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	94035-1000	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	Directed Research
Start Date:	10/01/2010	**End Date:**	09/30/2013
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	None		
COI Name (Institution):	Ellis, Stephen (NASA Ames Research Center) Kaiser, Mary (NASA Ames Research Center)		
Grant/Contract No.:	Directed Research		
Performance Goal No.:			
Performance Goal Text:			
	The objective of the Directed Research Project (DRP) titled "Assessing and Mitigating the Impact of Transmission Delays on Teleoperations" is to examine coupled human-system performance in the presence of temporally varying communication delays between the human controller(s) and remotely operated robot(s), and to propose countermeasures for delay-induced performance decrements. This DRP will be conducted in support of the NASA Enabling Technology Development and Demonstration (ETDD) Program's Human Exploration Telerobotics (HET) Project, in which ground-station operators will control robotic assets on board the International Space Station (ISS). Specifically, the DRP will concentrate on the element led by NASA ARC that is focused on ground-to-orbit control of the "Synchronized Position, Hold, Engage, and Reorient Experimental Satellites" (SPHERES) free-flying robots on the ISS. The DRP will focus on the range of time delays encountered in the ground-based control of the robotic assets on ISS,		

Task Description:	ranging from 20-50 ms (effectively for line-of-sight communication) up to 2-10 s for multiple satellite ground-station relayed (Tracking and Data Relay Satellite System, or TDRSS, and associated ground network) communication, and, in particular, as delay instantaneously varies because of real-time changes in communication paths and data buffering. First, we will conduct human-in-the-loop (HITL) performance experiments using visual display of a dynamic simulation representative of a variety of SPHERES operations requiring different movement precision under this range of time-delay conditions. Next, we will examine HITL performance under these conditions employing mitigation techniques for short time delays such as prediction algorithms that generate compensatory in command signal lead and, for longer delays, predictive "feed-forward" graphical overlays that "look ahead" and provide a virtual view showing the future pose and location of the robot. The goal of the second of the studies is to understand the performance trades between these techniques in a wider variety of environmental and latency conditions than is usually achievable during in situ experimentation. Finally, based on these empirical HITL results, we will design and test a strategy for combining and gracefully switching between mitigation techniques as telerobot system time delays vary across the millisecond to second range. To conduct the DRP studies, we will build our experiment testbed derived from elements of the HET ground-to-orbit SPHERES task, encompassing ground operator user interfaces as well as computer-based simulations of the SPHERES robots and ground-ISS communication links. This strategy allows us to run HITL tests that will reduce the operating environment to offer sufficient flexibility and control for human performance experiments, yet still maintain salient features of the HET tasks key for face-validity and applicability of the results. The results from our experimental studies will help define more focused and scientifically revealing experiments that could subsequently be conducted on the ISS. The aims of the proposed work are: 1) to employ human-in-the-loop (HITL) testing to empirically investigate the impact of variable communication delays, with latencies spanning from tens of milliseconds up to approximately five seconds, on coupled human-system performance for telerobotic systems; 2) to evaluate empirically the efficacy of existing time delay compensation schemes for this range of latencies for telerobotic tasks and control modes that have different required movement precision levels; and 3) to use the data resulting from these studies to identify the trade points between latency compensation schemes as a function of time delay and required task precision and then design and test strategies for gracefully switching between mitigation techniques as telerobot system time delays vary. The guidelines, tools, mitigation techniques and performance metrics developed from this research will help provide a rational basis for the design of teleoperation tasks to be carried out in the presence of communication delays. These products will in turn assist subsequent task, technology design, and validation experiment decisions regarding acceptable or desirable delay compensation techniques and define at what point to engage more autonomous operational modes.
Rationale for HRP Directed Research:	This research is directed because it contains highly constrained research, which requires focused and constrained data gathering and analysis that is more appropriately obtained through a non-competitive proposal.
Research Impact/Earth Benefits:	Potential Earth benefit from this work is for time-delayed teleoperation via the Internet or space satellite communication networks. Domains to which results of this work may ultimately be applied include telesurgery and remote piloting of unmanned autonomous vehicles (UAVs).
Task Progress:	Since notification of authority to proceed on March 4, 2011, we have developed an experimental simulation to empirically investigate human performance in a three-dimensional telerobotics positioning task. We plan to use the simulation software first in an examination of the performance impacts of time delay and later for compensation techniques to mitigate the effects of these delays. We have also supported our stakeholder, the SPHERES (for Synchronized Position Hold Engage Re-orient Experimental Satellites) element of NASA's ETDD (Enabling Technology Development and Demonstration) Program's HET (Human Exploration Telerobotics) Project, in preparation for their planned measurements of ground-to-ISS (International Space Station) communication delays and bandwidth that will inform our subsequent latency and compensation studies. In order to better understand teleoperator system users' abilities to operate under long time delays and, ultimately, to enable examination of compensation techniques for these time delays, we have begun to analyze the robustness of the classical finding that for delays above ~250 ms operators can no longer maintain smooth coordinated manual control but instead resort to a move-and-wait strategy. To test the generality of this finding and, in particular, to investigate the invariance of the critical latency associated with the onset of move-and-wait behavior, we have identified a quantitatively modulated stressor that appears to directly impact the difficulty of a manually driven teleoperation task. The stressor entails interposing a three-axis rotational misalignment between the visual display and manual input coordinates. This stressor is particularly useful because it can metrically introduce a wide range of task difficulty levels ranging from almost subliminal to those that make smoothly coordinated three-dimensional manual control almost impossible. We have chosen rotational misalignment as a stressor because it represents a problem commonly encountered in teleoperation: that of the remote camera not being optimally aligned with the operator's input coordinates. The specific task we have selected for this investigation is a three-dimensional Fitts-like task in which the participants must move a cursor from a central starting location to touch a variably sized sphere target. This type of task could serve as the basis for an interaction technique or for a motion-planning tool that could support the HET project's SPHERES robot satellite element with which we are collaborating. We have recently completed data collection with twelve participants in a first study (without time delay) to characterize the degree of control difficulty our stressor can introduce. Data analysis is currently in progress. We are beginning pilot investigations to outline the range of latencies that we will employ in our next study in which we will determine whether the point at which users switch to move-and-wait strategy depends upon task difficulty set by the rotational misalignment stressor. Additionally, we plan to identify a second transition point after which even a move-and-wait approach will no longer work. We believe such a second threshold could be associated with limitations of the users' short-term memory. The intent of this upcoming study is to help quantify the upper and lower time-delay bounds for which a teleoperation

	system user would naturally adopt a move-and-wait strategy and to understand the degree to which these bounds vary with task difficulty. These bounds will also guide the design of delay compensation experiments that we plan to conduct during FY13. Because MIT's camera and processor upgrade to the SPHERES robots will not be launched to ISS this fiscal year, our stakeholder, the HET-SPHERES team elected instead to consider the NEXUS-S Android smartphone camera (with cellular components disabled) as a provisional solution to provide video and real-time data feedback from ISS. The smartphone was launched on STS-135. We assisted the HET-SPHERES team in pre-flight safety testing of the phones. The HET-SPHERES team plans to use this smartphone to quantify both the bandwidth and time delay characteristics of various elements making up the ground-to-ISS communication network. The delay measurements they make will help inform the design of our time delay studies and validate the significance of our future experiment results.
Bibliography Type:	Description: (Last Updated: 04/13/2017)
Papers from Meeting Proceedings	Adelstein BD, Ellis SR, Kaiser MK. "Teleoperation in the presence of time delay." To be presented at Fourth IEEE International Conference on Space Mission Challenges for Information Technology, Palo Alto, Calif., August 2-4, 2011. 2011 IEEE Fourth International Conference on Space Mission Challenges for Information Technology (SMC-IT 2011). , Aug-2011